Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To learn the Basics of Machine Learning**	
Experiment No: 01	**Date:**	**Enrollment No: 92410133012**

Aim: To learn the Basics of Machine Learning

IDE: Google Colab

Theory:

Machine learning is a subset of artificial intelligence in the field of computer science that often uses statistical techniques to give computers the ability to "learn" (i.e., progressively improve performance on a specific task) with data, without being explicitly programmed. In the past decade, machine learning has given us self-driving cars, practical speech recognition, effective web search, and a vastly improved understanding of the human genome.

Machine learning tasks

Machine learning tasks are typically classified into two broad categories, depending on whether there is a learning "signal" or "feedback" available to a learning system:

- **Supervised learning:** The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs. As special cases, the input signal can be only partially available, or restricted to special feedback:
- **Semi-supervised learning:** the computer is given only an incomplete training signal: a training set with some (often many) of the target outputs missing.
- **Active learning:** the computer can only obtain training labels for a limited set of instances (based on a budget), and also has to optimize its choice of objects to acquire labels for. When used interactively, these can be presented to the user for labeling.
- **Reinforcement learning:** training data (in form of rewards and punishments) is given only as feedback to the program's actions in a dynamic environment, such as driving a vehicle or playing a game against an opponent.
- **Unsupervised learning:** No labels are given to the learning algorithm, leaving it on its own to find structure in its input. Unsupervised learning can be a goal in itself (discovering hidden patterns in data) or a means towards an end (feature learning).

Machine learning applications

In classification, inputs are divided into two or more classes, and the learner must produce a model that assigns unseen inputs to one or more (multi-label classification) of these classes. This is typically tackled in a supervised manner. Spam filtering is an example of classification, where the inputs are email (or other) messages and the classes are "spam" and "not spam". In regression, also a supervised problem, the outputs are continuous rather than discrete.

In clustering, a set of inputs is to be divided into groups. Unlike in classification, the groups are not known beforehand, making this typically an unsupervised task. Density estimation finds the distribution of inputs in

some space. Dimensionality reduction simplifies inputs by mapping them into a lower- dimensional space. Topic modeling is a related problem, where a program is given a list of human language documents and is tasked with finding out which documents cover similar topics.

Machine learning Approaches

Decision tree learning: Decision tree learning uses a decision tree as a predictive model, which maps observations about an item to conclusions about the item's target value. Association rule learning Association rule learning is a method for discovering interesting relations between variables in large databases.

Artificial neural networks

An artificial neural network (ANN) learning algorithm, usually called "neural network" (NN), is a learning algorithm that is vaguely inspired by biological neural networks. Computations are structured in terms of an interconnected group of artificial neurons, processing information using a connectionist approach to computation. Modern neural networks are non-linear statistical data modeling tools. They are usually used to model complex relationships between inputs and outputs, to find patterns in data, or to capture the statistical structure in an unknown joint probability distribution between observed variables.

Deep learning

Falling hardware prices and the development of GPUs for personal use in the last few years have contributed to the development of the concept of deep learning which consists of multiple hidden layers in an artificial neural network. This approach tries to model the way the human brain processes light and sound into vision and hearing. Some successful applications of deep learning are computer vision and speech recognition.

Inductive logic programming

Inductive logic programming (ILP) is an approach to rule learning using logic programming as a uniform representation for input examples, background knowledge, and hypotheses. Given an encoding of the known background knowledge and a set of examples represented as a logical database of facts, an ILP system will derive a hypothesized logic program that entails all positive and no negative examples. Inductive programming is a related field that considers any kind of programming languages for representing hypotheses (and not only logic programming), such as functional programs.

Support vector machines

Support vector machines (SVMs) are a set of related supervised learning methods used for classification and regression. Given a set of training examples, each marked as belonging to one of two categories, an SVM training algorithm builds a model that predicts whether a new example falls into one category or the other.

Clustering

Cluster analysis is the assignment of a set of observations into subsets (called clusters) so that observations within the same cluster are similar according to some pre designated criterion or criteria, while observations drawn from different clusters are dissimilar. Different clustering techniques make different assumptions on the

structure of the data, often defined by some similarity metric and evaluated for example by internal compactness (similarity between members of the same cluster) and separation between different clusters. Other methods are based on estimated density and graph connectivity. Clustering is a method of unsupervised learning, and a common technique for statistical data analysis.

Bayesian networks

A Bayesian network, belief network or directed acyclic graphical model is a probabilistic graphical model that represents a set of random variables and their conditional independencies via a directed acyclic graph (DAG). For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases. Efficient algorithms exist that perform inference and learning.

Reinforcement learning

Reinforcement learning is concerned with how an agent ought to take actions in an environment so as to maximize some notion of long-term reward. Reinforcement learning algorithms attempt to find a policy that maps states of the world to the actions the agent ought to take in those states. Reinforcement learning differs from the supervised learning problem in that correct input/output pairs are never presented, nor sub-optimal actions explicitly corrected.

Similarity and metric learning

In this problem, the learning machine is given pairs of examples that are considered similar and pairs of less similar objects. It then needs to learn a similarity function (or a distance metric function) that can predict if new objects are similar. It is sometimes used in Recommendation systems.

Genetic algorithms

A genetic algorithm (GA) is a search heuristic that mimics the process of natural selection, and uses methods such as mutation and crossover to generate new genotype in the hope of finding good solutions to a given problem. In machine learning, genetic algorithms found some uses in the 1980s and 1990s. Conversely, machine learning techniques have been used to improve the performance of genetic and evolutionary algorithms.

Rule-based machine learning

Rule-based machine learning is a general term for any machine learning method that identifies, learns, or evolves "rules" to store, manipulate or apply, knowledge. The defining characteristic of a rule-based machine learner is the identification and utilization of a set of relational rules that collectively represent the knowledge captured by the system. This is in contrast to other machine learners that commonly identify a singular model that can be universally applied to any instance in order to make a prediction. Rule-based machine learning approaches include learning classifier systems, association rule learning, and artificial immune systems.

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Feature selection approach

Feature selection is the process of selecting an optimal subset of relevant features for use in model construction. It is assumed the data contains some features that are either redundant or irrelevant, and can thus be removed to reduce calculation cost without incurring much loss of information. Common optimality criteria include accuracy, similarity and information measures.

Pre Lab Exercise:

a. Write 5 applications, that you encounter daily, where Machine Learning is used?

1. **Email Spam Detection** – Email services like Gmail automatically detect spam emails and move them to the spam folder using Machine Learning.
2. **Movie and Video Recommendations** – Platforms like Netflix and YouTube suggest movies or videos based on what we watched before. This recommendation system uses Machine Learning.
3. **Voice Assistants** – Voice assistants such as Google Assistant or Siri understand our voice commands and respond accordingly with the help of Machine Learning.
4. **Google Maps Navigation** – Google Maps predicts traffic conditions and suggests the best route using Machine Learning algorithms.
5. **Online Shopping Suggestions** – Shopping websites recommend products based on our previous searches and purchases using Machine Learning.

b. What do you mean by Supervised Learning?

Supervised Learning is a type of Machine Learning in which the model is trained using **labeled data**. This means that the input data already has the correct output associated with it. The algorithm learns the relationship between input and output and then uses that knowledge to make predictions on new data.

For example, if we want to predict house prices, the model is trained using past data that contains house features and their actual prices.

c. What do you mean by Unsupervised Learning?

Unsupervised Learning is a Machine Learning technique where the data **does not contain labeled outputs**. The algorithm analyzes the data and tries to find hidden patterns, similarities, or groupings within the dataset.

For example, grouping customers based on their purchasing behavior in a shopping store is an application of unsupervised learning.

d. What do you mean by Reinforcement Learning?

Reinforcement Learning is a type of Machine Learning where a system learns by **interacting with its environment**.
The system performs actions and receives **rewards or penalties** depending on whether the action is correct or incorrect. Over time, the system learns which actions give the best results.

A common example of reinforcement learning is **game-playing AI** or **self-driving cars**, where the system continuously learns from experience.

Program (Code):

Perform the following tasks:

1. Load a dataset in your IDE
2. Observe the statistics of all the features
3. Obtain the shape of the dataset
4. Separate all the features
5. Fill the missing values, if any, using the statistically relevant value
6. Observe the Box-Plot of each feature
7. Comment on the IQR range and outliers

Code :-

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Load Dataset
try:
    data = pd.read_csv("dataset.csv")
except FileNotFoundError:
    print("Error: 'dataset.csv' not found. Please upload the file to your Colab environment or provide the correct
      path.")
    print("Generating a dummy dataset for demonstration purposes to allow the rest of the code to run.")
    # Generate a dummy dataset
    np.random.seed(42) # for reproducibility
    dummy_data = {
        'Feature1': np.random.normal(loc=10, scale=2, size=100),
        'Feature2': np.random.normal(loc=50, scale=10, size=100),
        'Feature3': np.random.uniform(low=0, high=100, size=100),
```

```
    'Feature4': np.random.randint(low=1, high=10, size=100)
}
data = pd.DataFrame(dummy_data)
# Add some missing values for demonstration of fillna
data.loc[data.sample(frac=0.1).index, 'Feature1'] = np.nan
data.loc[data.sample(frac=0.05).index, 'Feature2'] = np.nan
data.loc[data.sample(frac=0.02).index, 'Feature3'] = np.nan

# 2. Observe statistics of all features
print("Statistics of Dataset")
print(data.describe())

# 3. Obtain shape of dataset
print("Shape of dataset:", data.shape)

# 4. Separate all features
features = data.iloc[:, :]
print("Features:")
print(features.head())

# 5. Fill missing values using mean
data.fillna(data.mean(), inplace=True)

# 6. Observe boxplot of each feature
data.boxplot(figsize=(10,6))
plt.title("Box Plot of Features-D12")
plt.show()

# 7. Comment on IQR and Outliers
Q1 = data.quantile(0.25)
Q3 = data.quantile(0.75)
IQR = Q3 - Q1

print("IQR values:")
print(IQR)
```

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```
# Detect Outliers
outliers = ((data < (Q1 - 1.5 * IQR)) | (data > (Q3 + 1.5 * IQR)))
print("Possible Outliers:")
print(outliers.sum())
```

Results:

To be attached with

```
... Generating a dummy dataset for demonstration purposes to allow the rest of the code to run.
    Statistics of Dataset
            Feature1   Feature2   Feature3    Feature4
    count  90.000000  95.000000  98.000000  100.000000
    mean    9.709453  50.306295  49.271938    4.990000
    std     1.828831   9.354181  27.392082    2.622725
    min     4.760510  30.812288   1.439349    1.000000
    25%     8.731467  42.072347  28.158574    3.000000
    50%     9.554552  50.996514  52.679895    5.000000
    75%    10.932920  55.543993  70.235478    7.000000
    max    13.704556  77.201692  97.585208    9.000000
    Shape of dataset: (100, 4)
    Features:
        Feature1   Feature2   Feature3  Feature4
    0  10.993428  35.846293  87.737307         5
    1   9.723471  45.793547  74.076862         3
    2  11.295377  46.572855  69.701574         1
    3        NaN  41.977227  70.248408         1
    4   9.531693  48.387143  35.949115         8
```

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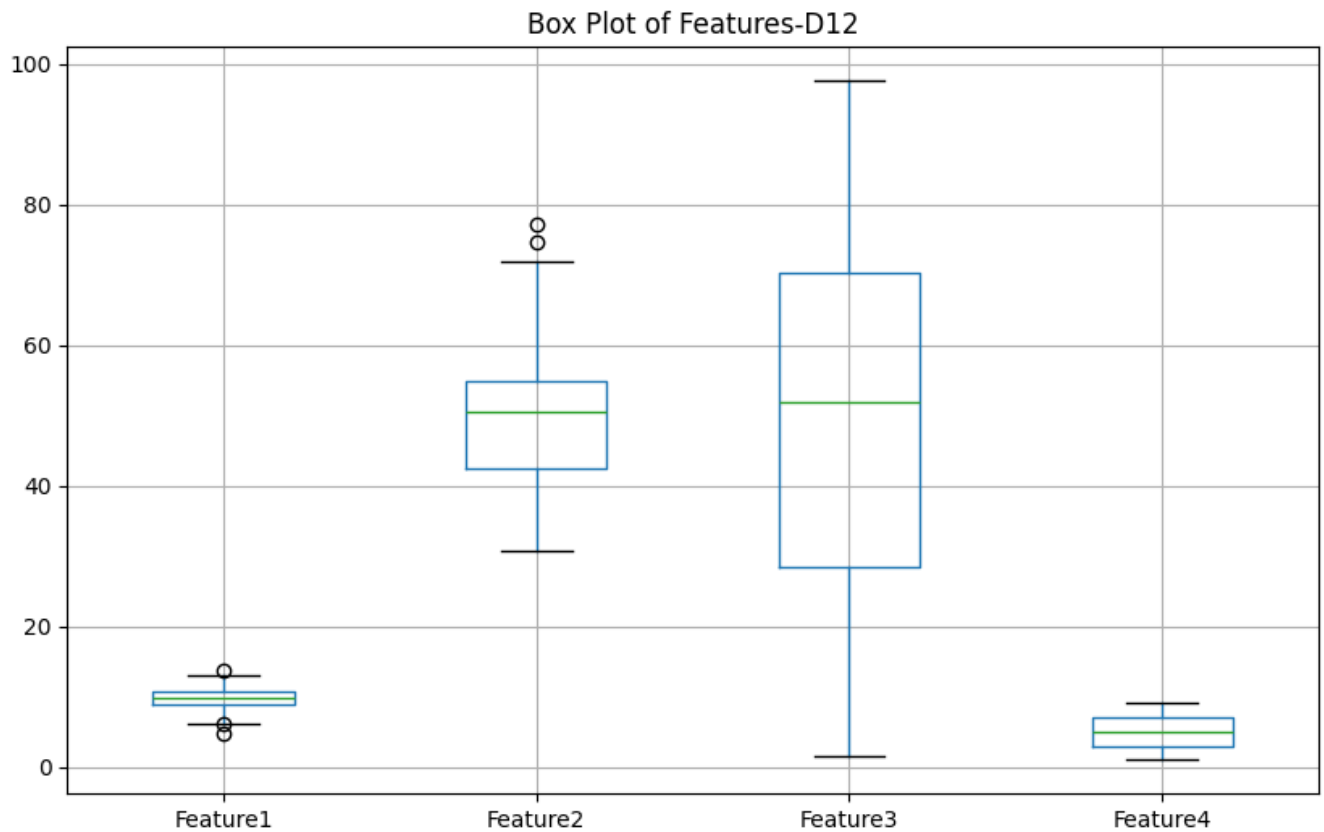


```
IQR values:
Feature1     1.866662
Feature2    12.534551
Feature3    41.720677
Feature4     4.000000
dtype: float64
Possible Outliers:
Feature1    3
Feature2    2
Feature3    0
Feature4    0
dtype: int64
```

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Observation and Result Analysis:

a. Statistics of the dataset

The statistical summary of the dataset provides useful information such as **mean, median, standard deviation, minimum value, maximum value, and quartile values** of each feature.
These statistics help us understand the **overall distribution and characteristics of the data**.

For example, the mean shows the average value of a feature, while the standard deviation indicates how much the data varies from the average.

b. IQR and Outliers

The **Interquartile Range (IQR)** is the difference between the third quartile (Q3) and the first quartile (Q1).

$$IQR = Q3 - Q1$$

It represents the **middle 50% of the dataset**.
Outliers are the values that lie far away from most of the data points. These can be detected using the following range:
Lower Limit = Q1 − 1.5 × IQR
Upper Limit = Q3 + 1.5 × IQR
Any value outside this range is considered an **outlier**.
Box plots are commonly used to visually identify these outliers in the dataset.

Post Lab Exercise:

a. Which is a better mean of replacing the missing value-mean, median or mode?

The best method depends on the type and distribution of the data.

- **Mean** is suitable when the data is normally distributed and does not contain extreme values.
- **Median** is better when the dataset contains outliers because it is not affected by extreme values.
- **Mode** is generally used for **categorical data**.

In most cases, **median is considered more reliable** because it is less sensitive to outliers.

b. How IQR range helps to understand the nature of the distribution of the data?

The IQR range helps us understand how the **middle portion of the data is spread**.
It shows how concentrated or spread out the central values of the dataset are.
A **small IQR** indicates that the data points are close to each other, while a **large IQR** indicates more variability in the data.
It is also useful for identifying **outliers** and understanding the overall distribution.

c. Which are the ways to replace the categorical data to the numerical ones?

Categorical data cannot be directly used in many Machine Learning algorithms, so it must be converted into numerical form. Some common methods are:

1. **Label Encoding** – Each category is assigned a unique number.
2. **One-Hot Encoding** – Separate columns are created for each category with binary values (0 or 1).
3. **Ordinal Encoding** – Used when categories have a natural order such as low, medium, and high.
4. **Binary Encoding** – Categories are converted into binary numbers to reduce dimensionality.